Practice Quiz: Lecture 4 (Part 1) — Digital Transmission & Line Coding

Topic: Digital Channels, Baseline Wandering, DC Components, Self-Synchronization, NRZ & Data Rates

Student Name: _____	Student ID: _____	Score: _____ / 20

> **Instructions:** Answer all 20 questions. For calculation questions, write intermediate steps in the provided work area. An answer key and explanations are provided at the end of the document.

MCQ **1. Which type of transmission medium is considered most suitable for direct baseband digital transmission according to the lecture?**

- ☐ a. A medium that passes only a single discrete frequency
- ☐ b. A low-pass medium
- ☐ c. A band-pass medium only
- ☐ d. A medium that blocks all low frequencies
- ☐ e. A carrier-modulated wireless channel only

MCQ **2. Why is a band-pass medium unsuitable for transmitting baseband digital signals without modulation?**

- ☐ a. It blocks high frequencies while allowing direct current (DC) to pass freely
- ☐ b. Baseband digital signals require low frequencies around zero, which are blocked by band-pass channels
- ☐ c. It doubles the bit rate of the incoming digital stream
- ☐ d. It eliminates all noise and prevents detection of signal amplitude
- ☐ e. It automatically forces the receiver clock to run faster than the sender clock

---

MCQ **3. Which condition is most likely to cause baseline wandering in a digital communication receiver?**

- ☐ a. A long sequence of alternating 0s and 1s
- ☐ b. Frequent voltage transitions across zero
- ☐ c. A long sequence of consecutive 0s or consecutive 1s
- ☐ d. Increasing the signaling baud rate
- ☐ e. Employing Manchester encoding

MCQ **4. What is the adverse effect of baseline wandering on signal decoding?**

- ☐ a. It permanently doubles the transmission bandwidth
- ☐ b. It causes the receiver's decision threshold to drift, making it difficult to distinguish 0s from 1s
- ☐ c. It converts digital pulses directly into frequency-modulated carrier waves
- ☐ d. It forces sender and receiver clocks to run perfectly in sync
- ☐ e. It removes high-frequency harmonics completely

MCQ **5. Why is the presence of a DC component undesirable in digital line transmission?**

□ a. It may be blocked by transmission media and components that do not pass very low frequencies, distorting the signal

□ b. It instantly makes the sender clock slower than the receiver clock

□ c. It eliminates the need for bandwidth in the transmission medium

□ d. It converts the digital signal into an analog voice wave

□ e. It decreases the total voltage amplitude to zero

MCQ **6. What gives a digital signal its self-synchronizing capability?**

□ a. Maintaining a constant non-zero voltage level across multiple bit intervals

□ b. Incorporating timing information through regular signal transitions within the transmission

□ c. Introducing a large constant DC voltage offset

□ d. Transmitting using only negative voltage polarities

□ e. Relying solely on an external dedicated clock wire without mid-signal transitions

---

MCQ **7. Which combination represents the most desirable characteristics for a digital transmission scheme?**

□ a. High DC component, no transitions, low bandwidth

□ b. Baseline wandering, DC component present, no synchronization

□ c. No baseline wandering, no DC component, and self-synchronization

□ d. Variable DC component, baseline drift, external clocking only

□ e. Fixed positive amplitude only, wide spectrum, absence of transitions

---

MCQ **8. What is the primary operational drawback associated with basic unipolar line encoding?**

□ a. It requires four distinct signal voltage levels

□ b. It has a nonzero average voltage amplitude, creating a persistent DC component

□ c. It requires two signal transitions per bit interval

□ d. It cannot represent binary 1 with a positive voltage level

□ e. It uses only alternating negative polarities

MCQ **9. Which statement correctly contrasts the synchronization vulnerabilities of NRZ-L and NRZ-I?**

□ a. NRZ-L loses synchronization on long runs of 0s or 1s, whereas NRZ-I loses synchronization primarily on long runs of 0s

□ b. NRZ-L has problems only with 0s, whereas NRZ-I has problems only with 1s

□ c. NRZ-I completely solves synchronization for all bit streams, whereas NRZ-L does not

□ d. Both NRZ-L and NRZ-I guarantee a transition in every single bit interval regardless of data

□ e. NRZ-L uses signal transitions for bit 1, while NRZ-I uses constant voltage levels for all data

---

True/False **10. True or False: In NRZ-I encoding, a long consecutive sequence of binary 1s causes frequent voltage transitions, thereby helping the receiver maintain bit synchronization.**

□ a. True

□ b. False

Calculation **11. A receiver clock is 0.2% faster than the sender clock. If the sender transmits data at a rate of $500\ \text{kbps}$, how many extra bits per second will the receiver detect?**

Calculation **12. A receiver clock is 0.15% slower than the sender clock. If the sender transmits data at a bit rate of $2\ \text{Mbps}$, how many bits are missed by the receiver each second?**

Calculation **13. An NRZ-I system transmits data at a bit rate of $N = 16\ \text{Mbps}$. Assuming a case factor $c = 0.5$ and $r = 1$ bit per signal element:**

(a) Calculate the average signal rate S in MBaud.

(b) Determine the minimum required channel bandwidth B_{min} in MHz.

Calculation **14. A line coding system transmits at a data rate of $N = 40\ \text{Mbps}$ with $c = 0.5$ and $r = 1$. Find the average signal rate S and the minimum bandwidth B_{min}.**

Calculation **15. A channel has an available bandwidth of $4\ \text{MHz}$. Assume $c = 0.5$ and each signal element carries $2\ \text{bits}$ ($r = 2$). Calculate the maximum achievable data rate N_{max} in Mbps.**

Calculation **16. A transmission channel has a bandwidth of $10\ \text{MHz}$. Assuming $c = 0.5$ and $r = 1\ \text{bit/symbol}$, calculate the maximum data rate N_{max} in Mbps that can be sent over this channel.**

Calculation **17. A digital transmission system achieves a data rate of $N = 24\ \text{Mbps}$ over a channel with bandwidth $B_{\min} = 6\ \text{MHz}$. Assuming an average case factor $c = 0.5$, how many bits does each signal element represent (r)?**

MCQ **18. Consider an NRZ-L transmission carrying the bit stream `00000000`. What is the fundamental problem observed at the receiver?**

- ☐ a. Extreme frequency dispersion across multiple bands
- ☐ b. Lack of voltage transitions leading to clock drift and baseline wandering
- ☐ c. Doubling of the Nyquist sampling rate
- ☐ d. Transition to high-frequency carrier wave modulation
- ☐ e. Overheating of transformers due to alternating high currents

True/False **19. True or False: If a digital signal contains a DC component, it can pass through AC-coupled transmission systems (such as those using transformers or coupling capacitors) without any signal distortion.**

- ☐ a. True
- ☐ b. False

MCQ **20. In signal rate calculations, the formula relating signal rate S, data rate N, case factor c, and bits per symbol r is:**

- ☐ a. $S = c \times N \times r$
- ☐ b. $S = \frac{c \times N}{r}$
- ☐ c. $S = \frac{N \times r}{c}$
- ☐ d. $S = \frac{c}{N \times r}$
- ☐ e. $S = \frac{r}{c \times N}$

Answer Key & In-Depth Explanations — Quiz 1

Question 1

Correct Answer: b. A low-pass medium

Explanation: A digital signal is a composite analog signal with infinite bandwidth starting from zero frequency (baseband). Hence, a low-pass medium (a channel passing frequencies starting from 0 Hz) is the most suitable medium for direct digital transmission[cite: 1].

Question 2

Correct Answer: b. Baseband digital signals require low frequencies around zero, which are blocked by band-pass channels

Explanation: Band-pass channels block lower frequencies near DC (0 Hz). Baseband digital signals have significant spectral energy concentrated around low frequencies, so sending them directly through a band-pass medium leads to severe attenuation and loss of data[cite: 1].

Question 3

Correct Answer: c. A long sequence of consecutive 0s or consecutive 1s

Explanation: The receiver assesses incoming signals by calculating a running average of received signal power (the baseline). A long string of consecutive 0s or consecutive 1s causes this average baseline to drift, a phenomenon known as baseline wandering[cite: 1].

Question 4

Correct Answer: b. It causes the receiver's decision threshold to drift, making it difficult to distinguish 0s from 1s

Explanation: In baseline wandering, the drifting baseline prevents the threshold detector from accurately determining whether a received voltage represents a binary 0 or 1, increasing decoding errors[cite: 1].

Question 5

Correct Answer: a. It may be blocked by transmission media and components that do not pass very low frequencies, distorting the signal

Explanation: When a signal has a non-zero average voltage, it exhibits a DC (Direct Current) component at 0 Hz. Components like transformers or AC coupling capacitors block frequencies close to zero, which truncates the signal and induces distortion[cite: 2].

Question 6

Correct Answer: b. Incorporating timing information through regular signal transitions within the transmission

Explanation: A self-synchronizing digital signal contains built-in transitions that allow the receiver's local clock to lock onto bit boundaries and resynchronize without needing an independent clock line[cite: 2].

Question 7

Correct Answer: c. No baseline wandering, no DC component, and self-synchronization

Explanation: The ideal criteria for a digital line coding scheme are: immunity to baseline wandering, zero DC component (permitting transformer coupling), and built-in self-synchronization to maintain receiver lock[cite: 2].

Question 8

Correct Answer: b. It has a nonzero average voltage amplitude, creating a persistent DC component

Explanation: In unipolar schemes (where one state is 0 V and the other is $+V$), the signal never dips below zero. Thus, its average amplitude is strictly positive, producing a large DC component[cite: 3].

Question 9

Correct Answer: a. NRZ-L loses synchronization on long runs of 0s or 1s, whereas NRZ-I loses synchronization primarily on long runs of 0s

Explanation: NRZ-L (Non-Return-to-Zero-Level) relies on constant voltage levels for both 0 and 1; thus, long strings of either bit produce no transitions. NRZ-I (Invert on 1s) transitions whenever a 1 occurs, meaning long strings of 1s provide frequent transitions, but long strings of 0s yield no transitions[cite: 3].

Question 10

Correct Answer: a. True

Explanation: In NRZ-I, every binary 1 forces an immediate inversion/transition in signal level. A continuous run of 1s generates transitions at every bit boundary, giving abundant timing information for the receiver clock[cite: 3].

Question 11

Correct Answer: 1,000 extra bits per second

Explanation:

$$\text{Clock Drift Error} = \text{Data Rate} \times \text{Drift Percentage}$$

$$\text{Extra Bits/s} = 500{,}000 \text{ bps} \times 0.002 = 1{,}000 \text{ extra bits/second}[\text{cite: 3}]$$

Question 12

Correct Answer: 3,000 bits per second

Explanation:

$$\text{Missing Bits/s} = 2{,}000{,}000 \text{ bps} \times 0.0015 = 3{,}000 \text{ bits/second}[\text{cite: 3}]$$

Question 13

Correct Answer: (a) S = 8 MBaud, (b) B_min = 8 MHz

Explanation: Using the signal rate formula:

$$S = c \times \frac{N}{r} = 0.5 \times \frac{16 \text{ Mbps}}{1} = 8 \text{ MBaud}[\text{cite: 4}]$$

The minimum theoretical bandwidth equals the average baud rate:

$$B_{\text{min}} = S = 8 \text{ MHz}[\text{cite: 4}]$$

Question 14

Correct Answer: S = 20 MBaud, B_min = 20 MHz

Explanation:

$$S = c \times \frac{N}{r} = 0.5 \times \frac{40 \text{ Mbps}}{1} = 20 \text{ MBaud}$$

$$B_{\text{min}} = S = 20 \text{ MHz}[\text{cite: 4}]$$

Question 15

Correct Answer: 16 Mbps

Explanation: Since the minimum bandwidth satisfies $B = S = 4$ MBaud, we rearrange $S = c \times \frac{N}{r}$:

$$N_{\text{max}} = \frac{1}{c} \times S \times r = \frac{1}{0.5} \times 4 \text{ MHz} \times 2 = 2 \times 4 \times 2 = 16 \text{ Mbps}[\text{cite: 4}]$$

Question 16

Correct Answer: 20 Mbps

Explanation:

$$N_{\text{max}} = \frac{1}{c} \times S \times r = \frac{1}{0.5} \times 10 \text{ MHz} \times 1 = 2 \times 10 = 20 \text{ Mbps}[\text{cite: 4}]$$

Question 17

Correct Answer: r = 2 bits/symbol

Explanation: Given $S = B_{\text{min}} = 6$ MBaud, $c = 0.5$, and $N = 24$ Mbps:

$$S = c \times \frac{N}{r} \implies 6 = \frac{0.5 \times 24}{r} = \frac{12}{r} \implies r = \frac{12}{6} = 2 \text{ bits/signal element}[\text{cite: 4}]$$

Question 18

Correct Answer: b. Lack of voltage transitions leading to clock drift and baseline wandering

Explanation: An unbroken run of binary 0s in NRZ-L holds a flat constant voltage. Without transitions, the receiver clock loses phase lock, and the baseline shifts, resulting in synchronization loss[cite: 1, 3].

Question 19

Correct Answer: b. False

Explanation: AC-coupled systems block frequencies at or near 0 Hz. Therefore, any digital signal possessing a DC component will experience severe attenuation of its low-frequency energy, inducing baseline distortion[cite: 2].

Question 20

Correct Answer: b. S = (c * N) / r

Explanation: The fundamental formula for baud rate is $S = c \times \frac{N}{r}$, where N is data rate in bps, r is bits per signal element, and c is the case factor (typically 0.5 for average cases)[cite: 4].
